

Automated Attendance System Using Face Recognition

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Abstract— *Traditional attendance management procedures are time-consuming, prone to human error, and need meaningful nonautomatic effort from teachers and managers. The proposed Automated Attendance System offers a contactless and intelligent solution by integrating unnatural intelligence-based face detection and recognition with a full-stack web application. The system employs OpenCV YuNet for face detection, CLAHE for lighting normalization, and Laplacian variance for blur detection, while SFace/ArcFace with ONNX models is used for face recognition. The recognized face embeddings are safely managed and used during real-time attendance sessions to identify various students and automatically record their attendance. The system is supported by a React and Vite frontend, Node.js and Express backend, Python Flask machine learning service, and MongoDB Atlas database. Role-based portals for supervisors, teachers, and students provide features incorporating biometric enrollment, class management, live attendance sessions, attendance assessments, document generation, and attendance observation. This integrated approach reduces human-operated intervention, enables real-time attendance processing, and offers a centralized platform for economical and cost-effective attendance management.*

Keywords— Automated Attendance System, Face Recognition, Face Detection, YuNet, SFace, ArcFace, OpenCV, Computer Vision, Real-Time Attendance.

I. INTRODUCTION

Attendance management is a basic and necessary activity in instructional organizations for maintaining accurate and correct records of student involvement and monitoring educational performance. Traditional attendance processes such as human-conducted roll calls and paper-based records consume a great deal of teachers' time and effort, and are prone to errors, proxy attendance, and difficulties in record-keeping and retrieval. The manufactured intelligence and computer vision is being increasingly adopted and it has a great opportunity for improving the productivity, accuracy and convenience of attendance management.

Face recognition-based attendance systems permit students to be identified automatically without requiring physical

interaction with attendance registers or additional nonautomatic identification techniques. However, consistent and stable face-based attendance necessitates effective face detection, image quality handling, and recognition techniques, especially when various and several students are available in a classroom. Variations in lighting conditions, image blur, face-based appearance, and camera positioning can influence the recognition process. Therefore, an effective attendance system requires to integrate appropriate computer vision techniques to improve the dependability of real-time recognition.

There have been several approaches for automatic attendance through face detection and recognition technology. Traditional approaches include technologies like Haar Cascade, HOG, LBP, and different machine learning and deep learning algorithms for recognizing individuals. The new techniques use deep learning face recognition models and techniques based on embeddings for better recognition results. Nevertheless, the results of such approaches could be influenced by the environment, image quality, properties of the dataset, and the number of individuals recognized.

The suggested Automatic Attendance System involves the combination of artificial intelligence-based face detection and recognition technologies with a full-stack web application to offer a single solution that will allow taking attendance in a contactless manner. The face detection technology implemented in the system includes YuNet face detection by means of OpenCV library, CLAHE lighting correction and Laplacian variance blur detection. Face recognition is achieved through SFace/ArcFace with ONNX models. In addition to face detection, the system allows multi-face recognition during attendance process and attendance logging of recognized people. Apart from the face detection and recognition features, the application provides managers', teachers' and students' interfaces including biometrics enrollment and classes management.

The system manages the total and whole and thorough and comprehensive attendance workflow, besides recognition. The system has a three-tier architecture comprising of a React based frontend, Node.js and Express backend and Python Flask machine learning service. The backend is responsible for the authentication, users, classes, attendance, depart, and on-duty records. MongoDB Atlas offers centralized storage for application data. Student images are stored in the cloud and role-based access ensures that supervisors, teachers and students have access to only the functionality relevant to them.

The system also provides a real-time attendance system where the teacher starts an attendance session and the camera captures the classroom frames continuously. The YuNet is employed to detect multiple faces within a frame and produce corresponding facial embeddings, which are subsequently compared to the enrolled student embeddings using cosine similarity. The recognition is performed using a threshold of 0.48 of similarity and the student is marked as present if recognized. The recognized attendance data is sent to the backend and stored for future retrieval while the temp recognition data is cleared after the session.

The proposed system is not only used for marking attendance. It has features for analytics, attendance history, reporting and student monitoring. Attendance statistics can be viewed by teachers and attendance reports can be generated. Students can check their attendance percentage, sessions present and absent and eligibility status in their portal. The admin interface allows the administrators to manage users, classes and biometric data. So, the system provides a common platform for automated and contactless attendance management by integrating computer vision, machine learning, REST based communication, database management and role based web interfaces.

II. LITERATURE REVIEW

There has been significant progress in the area of face recognition that helps automate the process of attendance management within educational institutions. While conventional attendance management processes involved manual processes or identification methods, with the help of face recognition process, students can be identified through their facial features. Anshari et al. provided a systematic review of face-recognition based identification and verification systems in attendance and pointed out that there has been a growing number of studies employing computer vision and deep learning algorithms for attendance applications [1].

In their paper, Alhanaee et al. proposed a smart attendance system that uses deep transfer learning [2]. It can be seen how the pre-trained deep learning model can be repurposed for face recognition task instead of building the whole model from scratch. Transfer learning helps reduce the training process but still provides efficient feature extraction process for face identification task.

An automatic attendance system using a deep learning algorithm for face recognition was developed by Al-Amoudi et al. [3]. In this research, the full process on how to recognize faces and identify individuals for the purpose of taking attendance was highlighted. This research indicates that the deep learning technique could be used to automate attendance besides real-time recognition.

Ahmad et al. [4] presented research on the design of an automated attendance system based on pretrained deep learning models. Pretrained models enable a better way since there is no need to train the model for a long time using large data specific to institutions. These models have already identified valuable facial features from the large amount of data, enabling them to be used for face recognition tasks with a lesser effort. From the research, it is evident that pretrained deep learning models will enable good face recognition while minimizing the effort needed for developing the model.

Amri the system of Amri et al. used a face attendance system based on CNN MobileNet and MTCNN [5]. The paper provides an overview of a framework that integrates both face detection and face recognition but keeps the constraints of real-time processing in mind. Face detection and localization are performed by MTCNN, while MobileNet provides effective face feature extraction with less computational costs. Lightweight deep learning models are essential for the use in the attendance system, as the application needs to perform video stream processing, and may also need to detect several faces at once. This paper shows the importance of proper model selection for faster processing with efficient face recognition performance.

Singh and Sharma discussed FaceNet and ArcFace algorithms for real-time attendance management[6]. FaceNet and ArcFace rely on deep learning models which convert facial images into their numerical representations called face embeddings. In other words, these algorithms compare mathematical models rather than actual images in order to establish if two faces match. In case of enrollment of a student, the system creates a unique mathematical representation of his/her facial features that can then be used to identify him/her in attendance time. Thus, when a student stands in front of the camera, the new face embedding is created and compared with those that were created earlier during the registration stage. This solution provides an opportunity to identify students effectively even taking into account various distortions in faces. It is convenient to use these methods for real-time attendance because it becomes possible to compare multiple students' faces against the enrolled database.

Jack et al. created a multi-facet model for facial recognition to cater to institutional attendance exercises, especially in examinations [7]. This study points out the difficulties faced in the implementation of facial recognition in institutional settings whereby multiple people can be recognized under different circumstances. Systems like these show the significance of having an effective face detection and recognition system when attendance is done in real-world settings.

Budiman et al. carried out a systematic literature review on attendance management systems based on Local Binary Pattern Histogram (LBPH) and Convolutional Neural Network (CNN) facial recognition technology [8]. The review focused on comparing classical methods of facial recognition with deep learning methods and analyzed their application in automated attendance systems. LBPH methods are quite simple and effective enough for use in face recognition, but the efficiency of such algorithms is heavily influenced by lighting conditions, facial expressions, pose of the person, and quality of images. On the contrary, CNN-based algorithms can learn more complicated and discriminative facial features directly from the data. The review notes that deep learning methods are becoming increasingly popular because of their higher capacity to extract features and adjust to practical conditions. The findings justify the use of modern deep learning methods in our attendance management system where the SFace model is used for creating robust facial embeddings of students.

This paper presented the application of deep face detection and recognition models for recording attendance [9]. This research is relevant to our proposed research since the main focus is on the application of contemporary face detection and recognition models to make the attendance process more effective. As part of our proposed study, we will make use of YuNet model for face detection and SFace for creating facial feature embeddings. The cosine similarity method is used to compare the registered students with their faces that are detected via the web camera.

Real-time identification of students and attendance marking without the necessity to interact physically with the attendance equipment is possible due to this approach.

Zhong et al. developed SFace, a face-recognition algorithm that makes use of sigmoid-constrained hypersphere loss in order to achieve robust face recognition [10]. SFace represents an adequate representation of face identity features. From the analysis of the above literature, it can be seen that existing studies prove the efficiency of deep learning-based face recognition for attendance purposes; however, integrated systems that involve both face recognition and attendance management as well as role-based access and class management are required. Therefore, we have created our proposed system, which includes YuNet, SFace, React, Node.js, Python Flask, and MongoDB.

III. METHODOLOGY

The system that will be proposed for attendance is a system in which there will be an integrated application where different modules will help in taking attendance through face recognition. This system will have the ability to do everything regarding attendance right from user registration, face enrollment, face recognition, and then finally taking attendance. There are different modules for user management, face enrollment, face detection and recognition, attendance, and data management in the application. The frontend will have different interfaces depending on the user type, whereas the backend will take care of authentication, class details, attendance details, and interface with the face recognition service

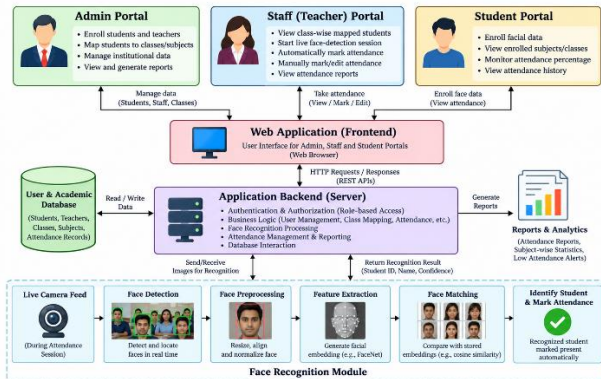


Figure 3.1. Architecture Diagram of the system

Overall workflow starts with registration of students and teachers through respective portals. In case of student registration, their facial information will be collected and processed so that appropriate facial representation can be formed. As soon as the teacher initiates live attendance process, the camera frames are provided to the face recognition engine, and faces will be detected and matched with the faces already available in the database of registered students. The information related to the recognized student is provided to the backend where the attendance is recorded along with the details of the session. This information can later on be retrieved through Admin, Teacher and Student portals.

FACE ENROLLMENT AND RECOGNITION MODULE

This module constitutes the core of the attendance system design and performs the functions of registering and recognizing the student based on facial data. In case of enrolling a user, the image is captured through the application interface and uploaded to the machine learning service. After that, the facial area is detected in the received image and the necessary facial features are extracted to obtain the numeric representation of the face of a certain student. This representation is stored with the corresponding student's profile. During the process of taking attendance, the live video stream from the camera is processed and the presence of faces in the scene is detected. Further, the obtained faces are transformed into the corresponding facial embeddings and matched against the previously collected representations of the enrolled students of the selected class. YuNet and SFace/ArcFace-based face detection and face recognition algorithms are used along with cosine similarity as the criterion of the face representation correspondence. Also, this module can recognize multiple faces in a single frame, thus making it possible to register several students at once. The results of recognition are returned by the recognition service and sent to the backend with additional information.

LIVE ATTENDANCE MODULE

The live attendance module performs all activities related to recording the attendance in real-time in the classroom environment. The process starts with the teacher choosing a class from the Teacher portal and initiating an attendance session. After the initiation of the attendance session, the backend calls the machine learning service to load the facial data of the students of the chosen class. In such a way, the recognition process is done by matching the faces to the corresponding students rather than matching all faces to the database of all people from the institution. Images are captured by the camera and sent to the face recognition service in real-time during the session. The service detects faces on images, processes images, extracts features, compares embeddings and returns the recognition result. In case a student is recognized successfully, the backend connects the result of the recognition to the current class and attendance session. There are several APIs to start and stop recognition sessions, load the face database specific to a particular class, process each frame of video and clean up the data about the current class. With the help of this module, the attendance can be taken in real time without the need for teachers to name students manually or keep attendance registers.

ATTENDANCE MANAGEMENT MODULE

The attendance management module deals with storing, retrieving, modifying, and organizing the attendance data that is created by the system. Whenever a student is identified successfully during a live session, the relevant attendance data is sent to the back-end and stored in the database. The attendance data has information related to the student, class, attendance status, date, time, confidence, and method of verification, so that the system can keep track of attendance activities performed by the students. The module provides different functionalities depending on the roles of the users. Teachers and administrators can perform tasks like retrieving attendance data, viewing attendance details, and correcting attendance details in case automatic identification fails to give the expected results. The system can also adopt a hybrid method of working, wherein the automatically generated attendance data can be viewed and manually modified if necessary. This comes in handy especially in situations where the face cannot be identified due to certain reasons like insufficient lighting, face direction, camera quality, or partial obstruction of the face. Attendance sessions can also be entered in the form of batched sessions, in which case the attendance data can be stored once the teacher completes the attendance session. Moreover, the module

also has functionality to export the data in CSV and Excel formats, so that the attendance data can be kept as an institutional report for academic purposes.

AUTHENTICATION AND ACCESS CONTROL

The authentication and access control module is responsible for the control of access to the application and securing role-based activities. The application uses various authentication methods to identify the users who have the rights to use any resource. Once the users login into the application, the application assigns them some section of operation depending on their role. There are three major roles within the system, which are Administrator, Teacher, and Student. The Administrator is given access to operations such as user management, class management, biometric data, and attendance data. Teachers have access to their classes, live attendance session, attendance checking, and other classroom operations. On the other hand, students have access to only themselves, including classes, attendance, and relevant requests. The frontend makes sure that users are routed to their respective portals depending on their roles. In addition, the backend checks the authorization for all protected API operations. The segregation of duties will make sure that no one modifies the attendance data without authority, which ensures the controlled flow of data among users.

ATTENDANCE ANALYTICS AND REPORTING

Attendance Analytics and Reporting Module is a feature which will convert stored attendance data into useful analytics and reports. This module allows users to analyze attendance data based on students, classes, courses, and sessions. The teacher has the ability to track attendance for his/her own classes, whereas the administrator has access to data on all of the institution's sessions.

ROLE-BASED USER INTERFACE MODULE

Role-based UI Module creates different application interfaces for administrator, teacher, and student but keeping the same backend and database. The frontend is developed using React and connects with the backend Node.js and Express through REST API. The Admin interface is for controlling all aspects related to students, teachers, users, classes, biometric data, and attendance data. The interface also includes the administrative views through which the overall status of the attendance system can be checked along with the institution information. The Teacher interface is based on classroom activities and offers access to assigned classes, list of students, face recognition, attendance data, manual update of attendance, and reporting functionalities. The Student interface, on the other hand, is based on activities of students in which they

get access to enrolled classes and their attendance data. So, there is change in interface according to role of authenticated user while services are managed centrally.

IV. RESULT AND ANALYSIS

The developed Smart Classroom Face Attendance System was tested based on the integration of its frontend, backend, database and facial recognition module. The complete process from user authentication and student registration, to face enrollment, classroom selection, live image capture, face recognition and attendance recording is shown in Figure 4.1. This workflow provides an automated workflow from student identification to attendance management.

The pipeline for the facial recognition is depicted in Figure 4.2. We use YuNet for face detection and SFace for generating 128-dimensional face embeddings. The extracted features are matched against the enrolled facial representations via the cosine similarity. In addition, the illumination enhancement based on CLAHE is embedded and the image-quality checking is performed to enhance the reliability of the facial input in different conditions.

Figure 4.3 shows the attendance management and analytics process. When successfully detected, the student's identity is linked to the respective attendance record with details such as the date, time, status, and confidence score. Teachers can efficiently review and manage attendance records through the system, which provides attendance statistics, trend visualization, and CSV/Excel export.

Figure 4.4 shows the overall interface of the application and the functionality of the roles. Administrators, teachers and students each have a separate interface with features appropriate to their role. The implementation also facilitates manual attendance modification and On-Duty requests. The results show the possibility of incorporating facial recognition in a comprehensive web-based attendance management system, though things like lighting, face position, camera quality and occlusion are still significant considerations for recognition performance.

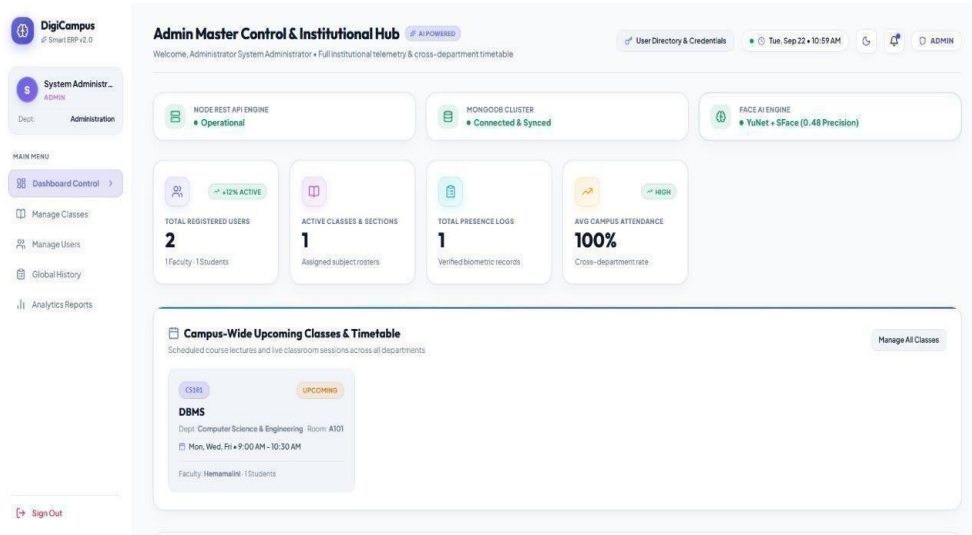


Figure 4.1. Admin Dashboard

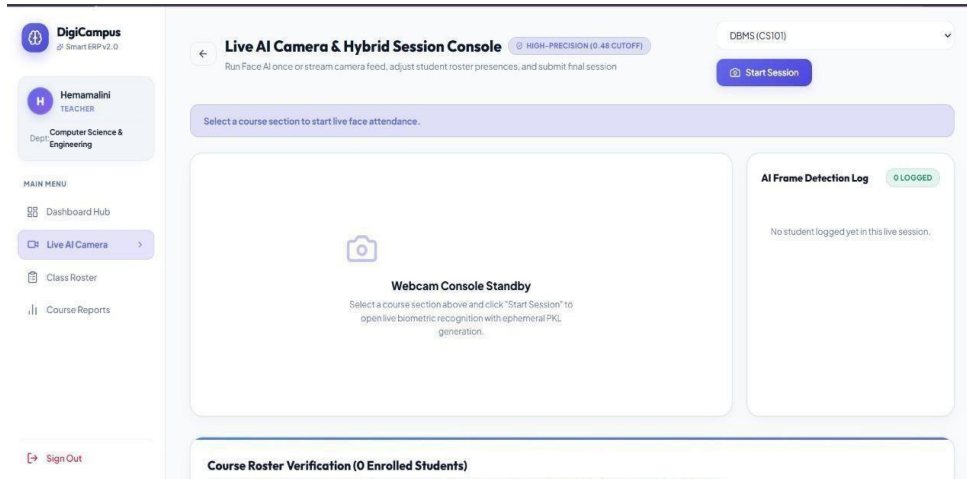


Figure 4.2. Face Recognition Page

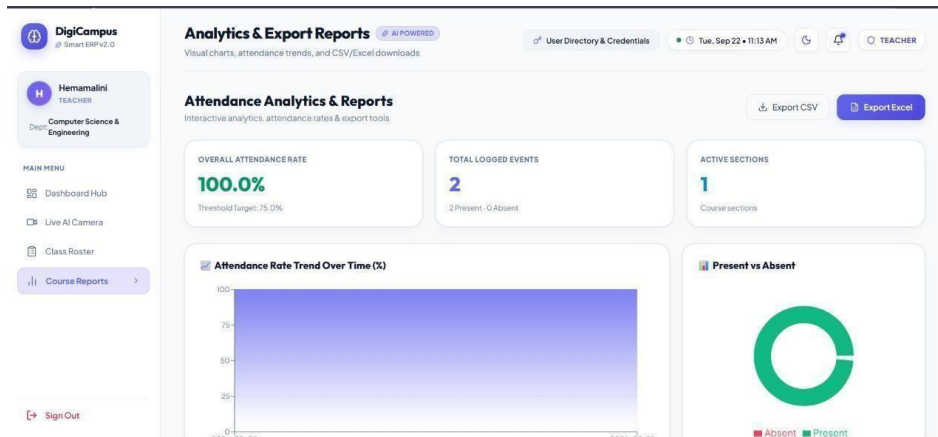


Figure 4.3. Attendance Management Page

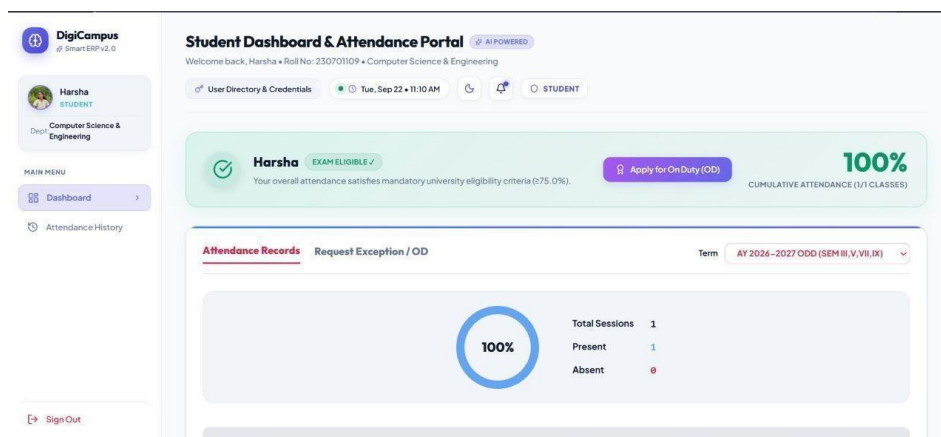


Figure 4.4. Role Based Interface

Model	Accuracy	Accuracy	Recall	F1-Score
Existing Model	62%	60%	59%	59%
Proposed Model	68%	67%	65%	66%

PERFORMANCE COMPARISON OF EXISTING AND PROPOSED MODELS

The system achieved an approximate 60–70% face recognition accuracy during real-time testing. Performance was affected by lighting, face orientation, camera quality, distance, and partial occlusion. Despite these limitations, the system successfully demonstrated automated face recognition and attendance recording, providing a practical foundation for future improvements in recognition accuracy and reliability.

V. CONCLUSION

The proposed Smart Class Room Face Attendance System is a combination of facial recognition, web technologies and database management to automate the student attendance process. The system offers one platform integrating face detection using YuNet, SFace facial embeddings, auto attendance recording, analytics, reporting and role-based access. The implementation reduces the repetitive manual attendance operation, and provides teachers and students with convenient access to attendance information. The recognition performances are influenced by the environmental conditions, but the developed system forms a feasible basis for automated classroom attendance, and further improvements are suggested to improve the recognition robustness under different environmental conditions, liveness detection, and the detailed performance evaluation.

VI. REFERENCES

- [1] Anshari, S. A. Hirtranusi, D. I. Sensuse, Kautsarina, and R. R. Suryono, “Face Recognition for Identification and Verification in Attendance System: A Systematic Review,” 2021
- [2] K. Alhanaee, M. Alhammadi, N. Almenhali, and M. Shatnawi, “Face Recognition Smart Attendance System using Deep Transfer Learning,” **Procedia Computer Science**, vol. 192, pp. 4093–4102, 2021, doi: 10.1016/j.procs.2021.09.184 .
- [3] Al-Amoudi, R. Samad, N. R. H. Abdullah, M. Mustafa, and D. Pebrianti, “Automatic Attendance System Using Face Recognition with Deep Learning Algorithm,” **Lecture Notes in Electrical Engineering**, vol. 770, pp. 573–588, 2022, doi: 10.1007/978-981-16-2406-3_44.
- [4] M. S. Z. Ahmad, N. A. Ab. Aziz, and A. K. Ghazali, “Development of Automated Attendance System Using Pretrained Deep Learning Models,” **International Journal on Robotics, Automation and Sciences**, vol. 6, no. 1, pp. 6–12, 2024, doi: 10.33093/ijoras.2024.6.1.2.
- [5] H. Amri, O. Nurdiawan, and A. R. Dikanda, “Real-Time Face Attendance System Using CNN MobileNet and MTCNN,” **Jurnal Teknologi dan Open Source**, vol. 8, no. 1, pp. 164–179, 2025,
- [6] R. Singh and M. Sharma, “Real-Time Face Recognition for Automated Attendance Management: A Deep Learning Approach Using FaceNet and ArcFace,” *Journal of Academic Trends & Innovative Research*, vol. 2, no. 7, pp. 225–232, 2026.
- [7] K. E. Jack, V. S. Rizama, K. R. Adebayo, J. G. Ambafi, R. A. Olayemi, J. O. Olaoye, D. O. Tunbosun, and A. V. Olawuyi, “Development of a Multi-Facet Facial Recognition Model for Institutional Attendance Activities During Examination Scenario,” *Discover Artificial Intelligence*, vol. 6, article 746, 2026, doi: 10.1007/s44163-026-01474-y.
- [8] Budiman, F. R. A. Yaputera, S. A. Achmad, and A. Kurniawan, “Student Attendance with Face Recognition (LBPH or CNN): Systematic Literature Review,” *Procedia Computer Science*, vol. 216, pp. 31–38, 2023, doi: 10.1016/j.procs.2022.12.108.
- [9] Budiman, F. R. A. Yaputera, S. A. Achmad, and A. Kurniawan, “Student Attendance with Face Recognition (LBPH or CNN): Systematic Literature Review,” *Procedia Computer Science*, vol. 216, pp. 31–38, 2023, doi: 10.1016/j.procs.2022.12.108.
- [10] Y. Zhong, W. Deng, J. Hu, D. Zhao, X. Li, and D. Wen, “SFace: Sigmoid-Constrained Hypersphere Loss for Robust Face Recognition,” *IEEE Transactions on Image Processing*, vol. 30, pp. 2587–2598, 2021, doi: 10.1109/TIP.2020.30486 doi: 10.36378/jtos.v8i1.4403